SCIENCEUTSAV®
Science is awesome

I'm a ROBO SCIENTIST

Light Sensing and Automatic Lighting Systems

Session

8

LIGHT SENSING AND AUTOMATIC LIGHTING SYSTEMS

Curiosity Kickoff

- How does a light sensor know whether it is day or night?
- What happens if you cover the light sensor with your hand?
- Why are automatic lights better than manual switches?

Concept Connection

This topic introduces students to Time Functionalities through the sessionLight Sensing and Automatic Lighting Systems. The goal is to keep the learning simple, practical, and beginner-friendly. Students first understand the concept in easy words, then see a small Arduino program, and then connect the code to the real output or input behavior.

1.Buzzer

A sound-making device that works when electricity makes a thin metal plate vibrate rapidly inside,The vibration creates buzzing or beeping sounds. Some buzzers can even change frequency to play simple tunes or melodies.

2 Watt SMD LED – The Super Torch

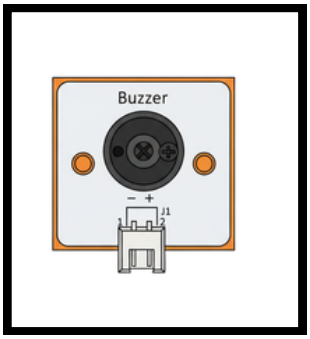
1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.

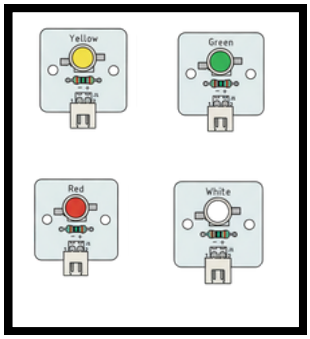
3. LDR Sensor :

An LDR sensor is a light-sensitive electronic component that changes its resistance according to the intensity of light falling on it. It is commonly used in automatic lighting systems, light detection circuits, security devices, and electronic projects to sense and respond to changes in ambient light**.**

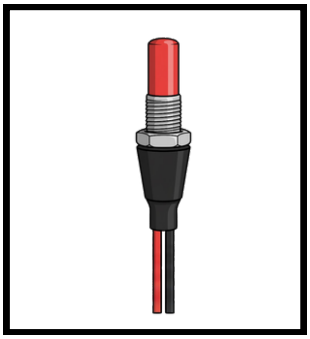
Push Button

A push button is a simple mechanical switch that controls an electrical circuit when pressed. It is widely used in electronic devices, control panels, automation systems, and DIY projects to provide user input and trigger specific actions.









Beginner: LDR Sensor Basics

Class Explanation

This topic introduces students to Time Functionalities through the session LDR and Laser Sensing. The goal is to keep the learning simple, practical, and beginner-friendly. Students first understand the concept in easy words, then see a small Arduino program, and then connect the code to the real output or input behavior.

Materials Required

- Arduino with attached shield
- 1 × LDR (Light Sensor)
- RMC wires
- USB cable

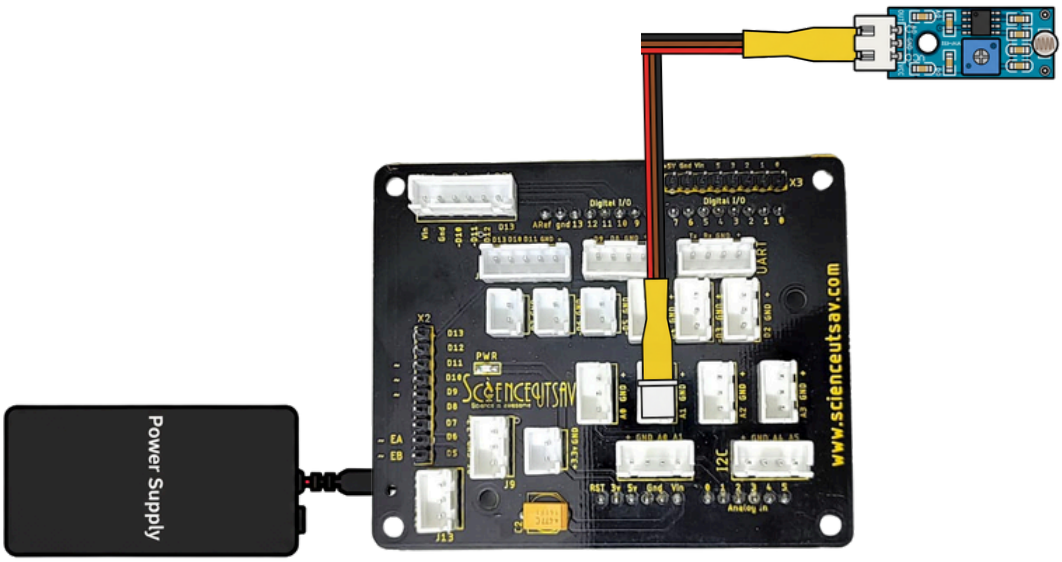
How to Connect

Use RMC wires to connect :

- 1 × LDR (Light Sensor) → Arduino pin A0 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
const int ldrPin = A1;

void setup() \
 Serial.begin(9600);
\

void loop() \
 int lightValue = analogRead(ldrPin);
 Serial.println(lightValue);
 delay(500);
\
```

Code Explanation

- In an Arduino program, first we define the pins or variables to give names to pin numbers so they are easy to use in the code. For example, we can define a sensor pin to read values.

- Next, the setup() function runs only one time when the Arduino starts. It is used to prepare the pins and start the Serial Monitor so we can see output on the screen.

- After that, the loop() function runs again and again continuously. This allows the Arduino to keep checking inputs and controlling the circuit without stopping.

- Inside the loop, analogRead() is used to read changing values from sensors like a potentiometer or LDR. These values keep changing instead of just ON or OFF. We can use Serial lines like Serial.println() to display these values in the Serial Monitor so we can understand what is happening.

So overall, setup() prepares everything once, and loop() keeps reading sensor values and displaying them continuously.

Try This Yourself Code

This Try This Yourself code keeps the same topic of LDR Sensor Basics but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int ldrPin = A1;
const int ledPin = 5;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 int lightValue = analogRead(ldrPin);

 if (lightValue < 500) \
   digitalWrite(ledPin, HIGH);
 \ else \
   digitalWrite(ledPin, LOW);
 \
\
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

the basic idea of LDR Sensor Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Intermediate: Automatic Light Control with LDR

Class Explanation

In this class, students learn the basic idea of Automatic Light Control with LDR. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

- Arduino with attached shield
- 1 × LDR (Light Sensor)RMC wires
- 1 × Red 1W SMD LED on breakout board
- USB cable

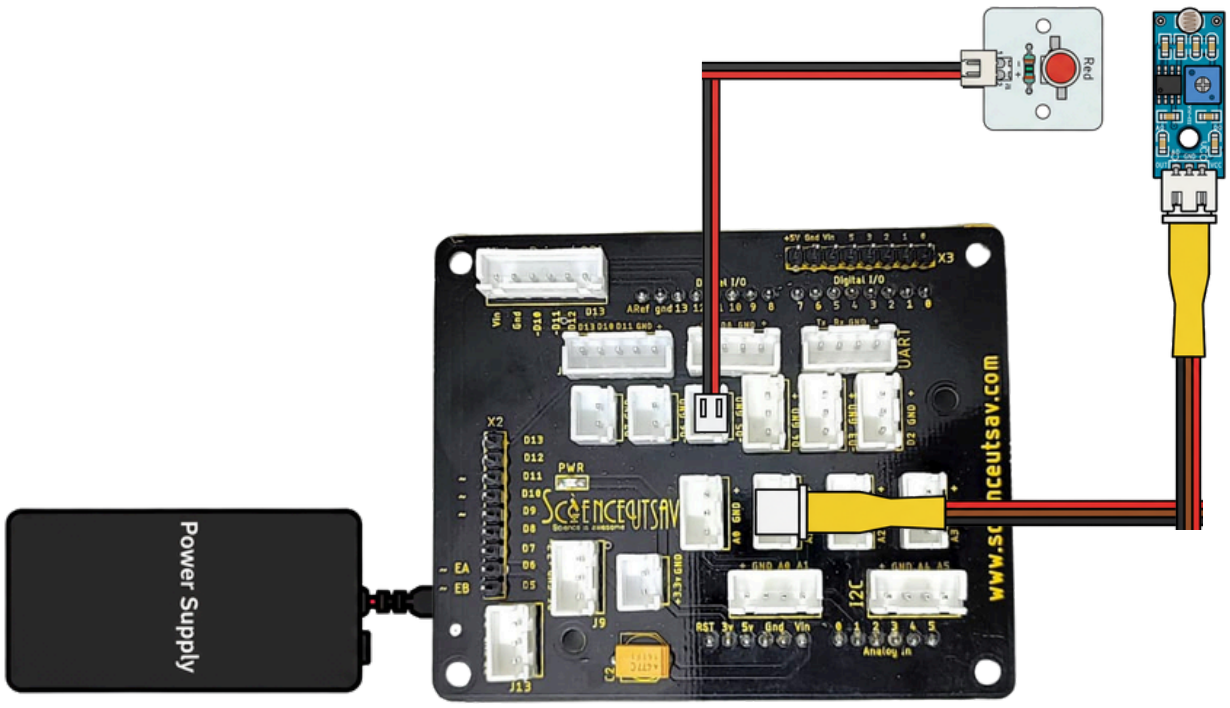
How to Connect

Use RMC wires to connect :

- 1 × LDR (Light Sensor)→ Arduino pin A1 and GND
- Red SMD LED → Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
const int ldrPin = A1;
const int ledPin = 5;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 int lightValue = analogRead(ldrPin);
 if (lightValue < 500) \
   digitalWrite(ledPin, HIGH);
 \ else \
   digitalWrite(ledPin, LOW);
 \
\
```

Code Explanation

- In an Arduino program, first we define the pins or variables to give names to pin numbers so they are easy to use in the code. For example, we can define a sensor pin and an LED pin.

- Next, the setup() function runs only one time when the Arduino starts. It is used to prepare the pins and start things like the Serial Monitor. Inside setup, we use pinMode() to tell whether a pin will work as an input or an output.

- After that, the loop() function runs again and again continuously. This allows the Arduino to keep checking inputs and controlling the circuit without stopping.

- Inside the loop, analogRead() is used to read changing values from sensors like a potentiometer or LDR. Based on these values, we can use an if condition to make decisions. For example, if the value is high, we can turn an LED ON, and if it is low, we can turn it OFF using digitalWrite().

So overall, setup() prepares everything once, and loop() keeps reading inputs and controlling outputs continuously using simple conditions.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Automatic Light Control with LDR but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int ldrPin = A1;
const int led1 = 5;
const int led2 = 6;

void setup() \
 pinMode(led1, OUTPUT);
 pinMode(led2, OUTPUT);
\

void loop() \
 int lightValue = analogRead(ldrPin);

 if (lightValue < 350) \
   digitalWrite(led1, HIGH);
   digitalWrite(led2, HIGH);
 \ else if (lightValue < 600) \
   digitalWrite(led1, HIGH);
   digitalWrite(led2, LOW);
 \ else \
   digitalWrite(led1, LOW);
   digitalWrite(led2, LOW);
 \
\
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:

the basic idea of Automatic Light Control with LDR how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Advanced: Laser LDR Beam Break Detection

Class Explanation

In this class, students learn the basic idea of Laser LDR Beam Break Detection. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

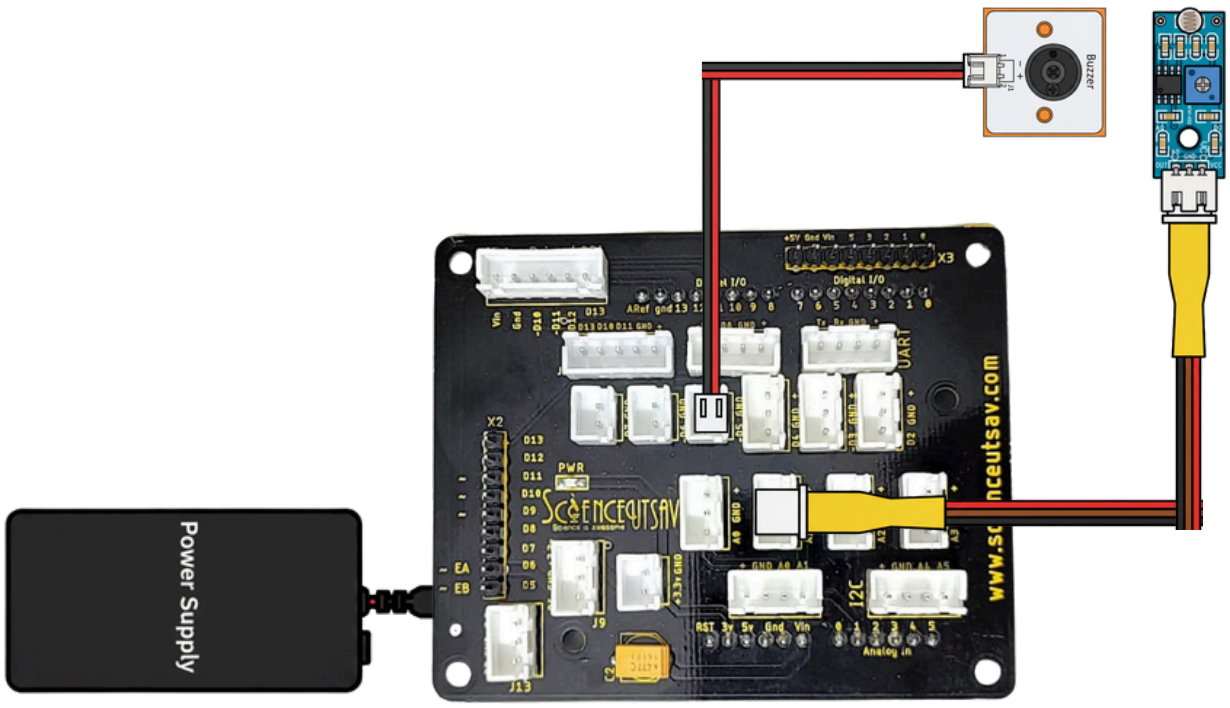
- Arduino with attached shield
- 1 × LDR (Light Sensor)
- 1 × Buzzer
- RMC wires
- USB cable

How to Connect

Use RMC wires to connect SMD LED breakout board directly to the Arduino:

- 1 × LDR (Light Sensor) → Arduino pin A1 and GND
- 1 × Buzzer → Arduino pin D5 and GND
- Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
const int ldrPin = A1;
const int buzzerPin = 7;

void setup() \
\

void loop() \
 int lightValue = analogRead(ldrPin);
 if (lightValue < 300) \
   tone(buzzerPin, 1000);
 \ else \
   noTone(buzzerPin);
 \
\
```

Code Explanation

- In an Arduino program, first we define the pins or variables to give names to pin numbers so they are easy to use in the code. For example, we can define a sensor pin and a buzzer pin.

- Next, the setup() function runs only one time when the Arduino starts. It is used to prepare the pins and start things like the Serial Monitor.

- After that, the loop() function runs again and again continuously. This allows the Arduino to keep checking inputs and controlling the circuit without stopping.

- Inside the loop, analogRead() is used to read changing values from sensors like a potentiometer or LDR. Based on these values, we can use an if condition to make decisions. For example, if the value is high, we can turn ON a buzzer using tone(), and if it is low, we can stop the sound using noTone().

So overall, setup() prepares everything once, and loop() keeps reading inputs and controlling devices like a buzzer using conditions.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Laser LDR Beam Break Detection but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int ldrPin = A1;
const int ledPin = 5;
const int buzzerPin = 7;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 int lightValue = analogRead(ldrPin);

 if (lightValue < 300) \
   digitalWrite(ledPin, HIGH);
   tone(buzzerPin, 1000);
 \ else \
   digitalWrite(ledPin, LOW);
   noTone(buzzerPin);
 \
\
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:
the basic idea of Laser LDR Beam Break Detection how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Expert: Sensor Fusion LDR and Touch

Class Explanation

In this class, students learn the basic idea of Sensor Fusion LDR and Touch. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

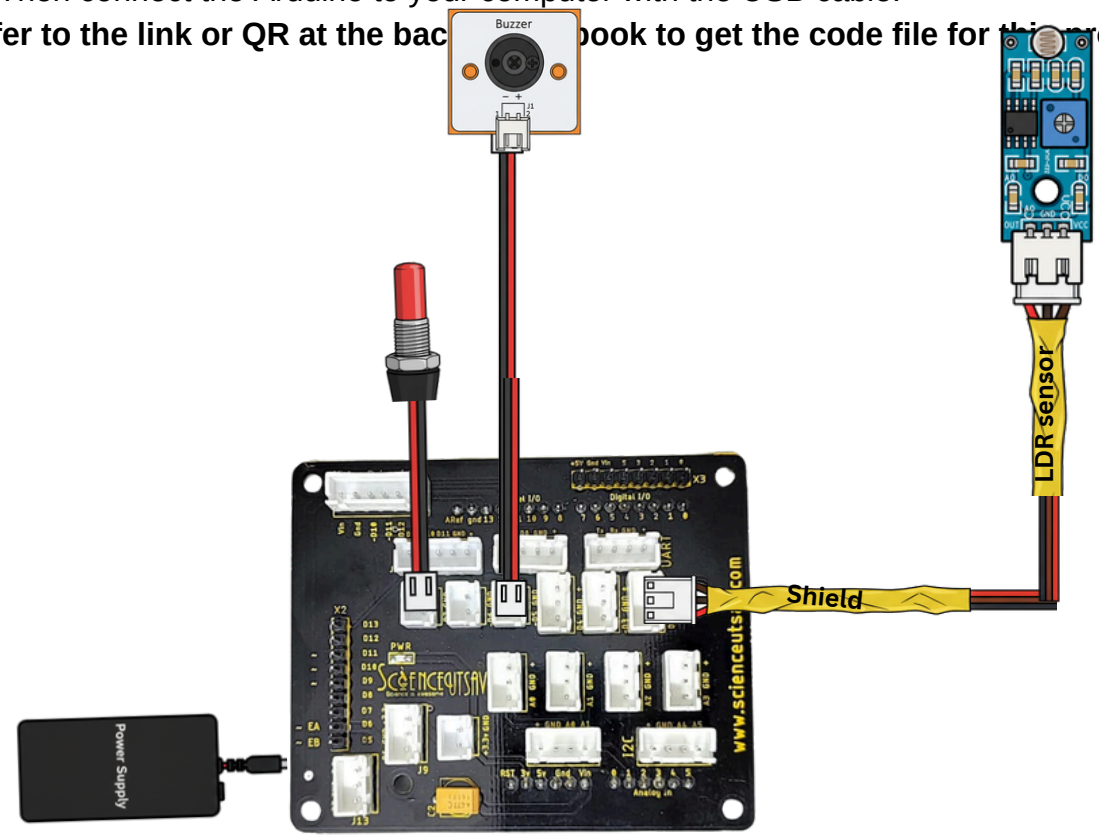
- Arduino with attached shield
- 1 × Red 1W SMD LED on breakout board
- 1 × LDR (Light Sensor)
- 1 × push Button
- Touch Sensor
- RMC wires
- USB cable

How to Connect

Use RMC wires to connect:

- 1 × LDR (Light Sensor) → Arduino pin A1 and GND
- 1 × Touch Sensor → Arduino pin D2 and GND
- 1 × push button
- Red SMD LED → Arduino pin D5 and GND
- Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the bac book to get the code file for this project.”



Code

```
const int touchPin = 2;
const int ldrPin = A1;
const int ledPin = 5;

void setup() \
 pinMode(touchPin, INPUT_PULLUP);
 pinMode(ledPin, OUTPUT);
\

void loop() \
 int lightValue = analogRead(ldrPin);
 bool touched = digitalRead(touchPin) == LOW;
 if (touched && lightValue < 500) \
   digitalWrite(ledPin, HIGH);
 \ else \
   digitalWrite(ledPin, LOW);
 \
\
```

Code Explanation

- In an Arduino program, first we define the pins or variables to give names to pin numbers so they are easy to use in the code. For example, we can define a button pin, a sensor pin, and an LED pin.

- Next, the setup() function runs only one time when the Arduino starts. It is used to prepare the pins and start things like the Serial Monitor. Inside setup, we use pinMode() to tell whether a pin will work as an input or an output.

- After that, the loop() function runs again and again continuously. This allows the Arduino to keep checking inputs and controlling the circuit without stopping.

- Inside the loop, digitalRead() is used to check the state of a button or switch (HIGH or LOW). We also use analogRead() to read changing values from sensors like a potentiometer or LDR. Based on these inputs, we use an if condition to make decisions. For example, if the button is pressed or the sensor value is high, we can turn an LED ON using digitalWrite(), otherwise we turn it OFF.

So overall, setup() prepares everything once, and loop() keeps reading inputs and controlling outputs continuously using simple conditions.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Sensor Fusion LDR and Touch but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int touchPin = 2;
const int ldrPin = A1;
const int ledPin = 5;
const int buzzerPin = 7;

void setup() \
 pinMode(touchPin, INPUT_PULLUP);
 pinMode(ledPin, OUTPUT);
\

void loop() \
 bool touch = digitalRead(touchPin) == LOW;
 int lightValue = analogRead(ldrPin);

 if (touch && lightValue < 400) \
   digitalWrite(ledPin, HIGH);
   tone(buzzerPin, 800);
 \ else \
   digitalWrite(ledPin, LOW);
   noTone(buzzerPin);
 \
\
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

the basic idea of Sensor Fusion LDR and Touch how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Learning Outcomes

After completing this session, students should be able to:ve:

- the basic idea of LDR Sensor Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketch
- the basic idea of LDR Sensor Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketchJ
- the basic idea of Laser LDR Beam Break Detection how the code connects with the circuit behavior how to upload and test a simple Arduino sketch
- the basic idea of Sensor Fusion LDR and Touch how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Try Out Other Things

- use the Try This Yourself code after each main class code before asking students to build a completely new variation
- ask students to point out the exact lines that changed between the first code and the Try This Yourself codecode
- let students first run the Try This Yourself code code as it is and then change one value, one condition, or one output